

Training the visual world model

PROPOSED SYSTEM

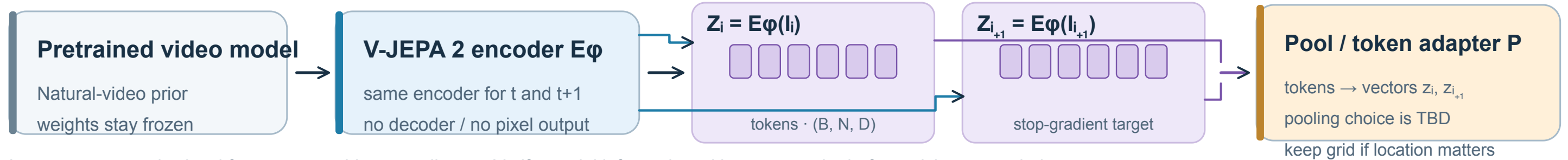
Module-level view · real simulator transitions become latent targets, then only D and R are optimized

01 DATASET · ONE ACTION-ALIGNED TRANSITION (synthetic MuJoCo domain)



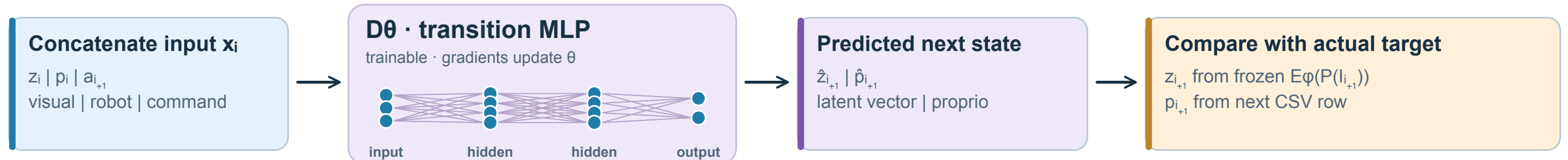
Collector saves post-action static + wrist RGB frames (256×256, 10 Hz), proprioception, action, goal and reward.

02 FROZEN VISUAL REPRESENTATION · ENCODE BOTH REAL OBSERVATIONS



Latents are numeric visual features, not object coordinates. Verify spatial information with a pose probe before relying on pooled z.

03 DYNAMICS MODEL D · SUPERVISED ONE-STEP PREDICTION



$$\text{LOSS } L_D = \lambda_z \cdot \text{MSE}(\text{norm}(\hat{z}_{i+1}), \text{norm}(z_{i+1})) + \lambda_p \cdot \text{MSE}(\hat{p}_{i+1}, p_{i+1})$$

Backpropagate through D only. $E\phi$ and target z_{i+1} are frozen / detached; normalize and weight proprio dimensions.

04 REWARD MODEL R · LEARN TO SCORE A PREDICTED OUTCOME

